

JOEL KALLARACKAL

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EDUCATION

Indian Institute of Technology, Madras

Bachelors of Technology, Engineering Physics

CGPA - 8.79/10

Oct 2022 – Present

PUBLICATIONS

- Hanlin Shi, **Joel Kallarackal**, Karthik Karumanchi, Sylvain Pellegrini, Andrew Orekhov, Andy Vu, Bhaskar Vundurthy, and Howie Choset. "A Unified Compliant Locomotion Framework for Snake Robot Navigation in Complex External Pipe Networks." Submitted to IEEE Transactions on Automation Science and Engineering (T-ASE).

RESEARCH EXPERIENCE

Motion Planning and Control of Snake Robots

Jul 2025 – Jul 2026

Prof. Howie Choset, Robotics Institute, Carnegie Mellon University

Pittsburgh, PA

Kinodynamic Motion Planning for a Modular Underwater Snake Robot

- Developed a motion-planning framework combining Interleaved Search and Trajectory Optimization (**INSAT**) with Graph of Convex Sets (**GCS**) for planning collision-free, dynamically feasible trajectories.
- Developed **dynamics model** incorporating joints, propeller thrust, gravity, buoyancy, and hydrodynamic drag.
- Developed an IMU-based attitude estimator using an error-state Kalman filter for robust orientation estimation.
- Implemented model based gravity compensation and designed PID-based station-keeping controller.
- Characterized the robot's stable configuration manifold, reducing the dimensionality of the planning problem.

Anchoring and Reaching Behavior for Confined Space Navigation

- Assembled and integrated a **16-link modular snake robot** with 90° joint offsets between consecutive links.
- Developed a **configurable testbed** to evaluate anchoring and reaching in confined spaces.
- Deployed and validated a kinodynamic motion-planning framework, demonstrating autonomous anchoring and reaching on the physical robot.

Complex Pipe Network Navigation

- Implemented **virtual chassis** body frame and developed **differential kinematics** for snake robot.
- Simulated **windowed rolling-helix** gait for T-junction navigation, and **spiraling** gait for obstacle avoidance.
- Developed a PyBullet simulation of contact-aware compliant control using torque sensing to adapt body curvature to varying pipe diameters.

Game-Theoretic Multi-Agent Ergodic Search

Oct 2025 – Mar 2026

Prof. Howie Choset, Robotics Institute, Carnegie Mellon University

Pittsburgh, PA

- Developed a game-theoretic multi-agent ergodic search framework by integrating Spectral Multiscale Coverage(SMC) with normal-form games to maximize coverage under adversarial conditions.
- Conducted **simulations** and **TurtleBot experiments** to validate multi-agent coverage and adversarial strategies.

Autonomous Ground Vehicle Development

Mar 2023 – Nov 2024

Prof. Prabhu Rajagopal, Prof. Sathyan Subbiah, Center for Innovation, IIT Madras

Chennai, India

Autonomous Electric Golf Cart

- Developed and deployed a ROS-based autonomy stack, transforming an electric golf cart into a driverless shuttle.
- Implemented a GPS waypoint navigation system for autonomous route planning.
- Fused depth data from stereo-camera with segmented non-drivable region from RGB image for costmap generation.
- **Accelerated** costmap generation of non-drivable region obtained via semantic segmentation **300x** by parallelization

Intelligent Ground Vehicle Competition (IGVC), Michigan, USA

- Contributed to an autonomous ground vehicle for IGVC, an international robotics competition.
- Designed a **robust navigation stack** for the robot as a distributed ROS system on a Raspberry Pi 4 and Jetson Orin
- Implemented a pure pursuit controller to manage lateral control and ensure **precise path tracking** for the vehicle.
- Devised an algorithm which **extracts inclines/ramps** from 3D point clouds utilizing DBSCAN for clustering.

Visuomotor Reinforcement Learning for Soft Tissue Manipulation

Dec 2023 – Apr 2024

Prof. Nirav Patel, Department of Engineering Design, IIT Madras

Chennai, India

- Developed a visuomotor reinforcement-learning agent for grasping target locations on deformable soft tissue.
- Reduced dimensionality of the observation space using convolutional autoencoders.
- Used SOFA framework to simulate soft-body dynamics and setup the RL environment.
- Generated a synthetic dataset for offline reinforcement learning.

Gaze Prediction from Lensless Camera Images

Dec 2024 – Apr 2025

Prof. Kaushik Mitra, Department of Electrical Engineering, IIT Madras

Chennai, India

- Developed a 3D gaze-estimation model from lensless images, achieving a 1.17° mean angular error.
- Reconstructed eye images from lensless measurements using a fine-tuned Dense Prediction Transformer (DPT).

Localization and Control for a Skid-Steering Mobile Robot

Jun 2024 – Aug 2024

Prof. Suresh Sundaram, Department of Aerospace Engineering, Indian Institute of Science, Bangalore

Bangalore, India

- Developed a kinematic model and velocity control interface for a skid-steering mobile robot.
- Tested and deployed LiDAR-inertial SLAM systems, including LIO-SAM and FAST-LIO.

COURSE PROJECTS

Port Hamiltonian Modelling and Control of Snakeboard

Jul 2025 – Nov 2025

Course: Geometry of Motion*, Prof. Howie Choset

- Derived **reduced momentum coordinates** spanning allowed momentum subspace of the non-holonomic system.
- Modelled the snakeboard as a **port hamiltonian system** and **simulated dynamics**.
- **Constructed bond graphs** which revealed the energy structure, providing insights for controller development.
- Developed a **energy based controller** which transfers energy from the torso to the forward momentum over time.

Multi Agent Planning with Deadlock Mitigation

Jan 2025 – May 2025

Course: Motion Planning, Prof. Nirav Patel

- Developed an A* variant with time as an additional planning dimension, enabling emergent waiting behavior.
- Handled diverse multi-robot deadlock scenarios that conventional A* planning fails to resolve.

Deep Learning with Differential Privacy

Jul 2024 – Nov 2024

Course: Introduction to Machine Learning, Prof. Sheetal Kalyani

- Integrated (ϵ, δ) -differential privacy in deep learning models to protect data confidentiality in computer vision tasks.
- Conducted experiments on MNIST and CIFAR-10 datasets, evaluating the impact of privacy-preserving techniques.
- Applied **Differentially Private SGD** with gradient perturbation, gradient clipping, and adaptive composition

TECHNICAL SKILLS

Programming Languages: Python, C++ , MATLAB

Software & Simulation: MuJoCo, PyBullet, Gazebo, CARLA, Solidworks, LTSpice, Mathematica

Robotics, Dynamics & Control: ROS, ROS 2, Pinocchio, CasADi, Crocoddyl, Drake

Other: PyTorch, Stable-Baselines3, OpenCV, Open3D, Git, Docker, Linux, LaTeX

RELEVANT COURSEWORK

Math: Differential Geometry, Probability, Linear Algebra, Information Theory, Optimization**

Robotics: Geometry of Motion*, Motion Planning, Linear Dynamical Systems**, Synthesis of Control**, Data Driven Control**, Optimal Control*

Learning: Introduction to Machine Learning, Deep Learning for Computer Vision, Reinforcement Learning.

Other: Computational Geometry, Classical Dynamics